

Kraft Basic Math Questions ✓

Part A SNT-TC-1A, CR189 and CR105-

Q7, 80% is minimum or practical.

$$\frac{24}{30} = .8 \text{ or } 80\%$$

Part B. (Metallurgy and Manufacturing)

Q 87. An order of magnitude is a factor of 10.

1 micrometer = 10^{-6} m or .000001 m

1 km = 10^3 m or 1000 m

.000001 to 1000

$$\frac{.000001}{1000} = \underline{\underline{9 \text{ places}}}$$

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Part C Methods

Q31

Inverse square Law:

$$I_1 D_1^2 = I_2 D_2^2$$

As distance increases intensity will decrease according to the distance squared. Intensity is inversely proportional to square of distance.

Example: $\frac{10 \text{ mR}}{\text{hr}} \times (3 \text{ m})^2 = I_2 \times (6 \text{ m})^2$

$$\frac{10 \text{ mR} \times 9 \text{ m}^2}{\text{hr}} = \frac{I_2 \times 36 \text{ m}^2}{36 \text{ m}^2}$$

$$2.5 \frac{\text{mR}}{\text{hr}} = I_2$$

Q38 (2) - (2T) - Hole size

Example: Part thickness = .5 inch

→ .02 x .5 inch = .01 inch, which is 1QT thickness

$$2T = 2 \times .01 \text{ inch} = .02 \text{ inch holes}$$

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Q 42

$$I_1 t_1 = I_2 t_2$$

$$50 \text{ ci} \times 10 \text{ minutes} = 20 \text{ ci} \times t_2$$

$$\frac{50 \text{ ci} \times \text{minutes} = 20 \text{ ci} \times t_2}{20 \text{ ci}} \quad \frac{20 \text{ ci} \times t_2}{20 \text{ ci}}$$

$$2.5 \text{ minutes} = t_2$$

Q 43

$$I_1 t_1 = I_2 t_2$$

Assume source initially had 100 ci or 100% activity.

100% initial 200% final

$$100 \text{ ci} / 50 \text{ ci} / 25 \text{ ci} / 12.5 \text{ ci}$$

If initial exposure for a 100 ci source is 1 minute, then:

$$100 \text{ ci} \times 1 \text{ min} = 12.5 \text{ ci} \times t_2$$

$$\frac{100 \text{ ci} \times 1 \text{ min} = 12.5 \text{ ci} \times t_2}{12.5 \text{ ci}} \quad \frac{12.5 \text{ ci} \times t_2}{12.5 \text{ ci}}$$

$$8 \text{ min} = t_2 \quad ; \quad 8 \text{ times longer}$$

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Q 44. 2 inches of lead is 4 half value layers.

$$2 \text{ inch Pb} \times \frac{1 \text{ HVL}}{.5 \text{ inch Pb}} = 4 \text{ HVLs}$$

$$\begin{array}{c|c|c|c} 100 \text{ mR/hr} & 50 \text{ mR/hr} & 25 \text{ mR/hr} & 12.5 \text{ mR/hr} \\ \hline 1 \text{ HVL} & 2 \text{ HVL} & 3 \text{ HVL} & 4 \text{ HVL} \end{array} \quad \frac{12.5 \text{ mR/hr}}{6.25 \text{ mR/hr}}$$

Q 47. Inverse square law (safety)

$$I_1 D_1^2 = I_2 D_2^2$$

$$200 \text{ mR/hr} \times (25 \text{ ft})^2 = I_2 \times (125 \text{ ft})^2$$

$$\frac{200 \text{ mR/hr} \times 625 \text{ ft}^2}{15625 \text{ ft}^2} = I_2 \times 15625 \text{ ft}^2$$

$$\frac{8 \text{ mR/hr}}{15625 \text{ ft}^2} = I_2$$

$$Q 81. \int = 26 \sqrt{\frac{1}{201} \text{ ft}}$$

Increasing frequency will result in a smaller number, therefore $\int$ (standard depth of penetration) will be less.

Part D (Miscellaneous) ^{5/10} KRAFT

Q25. $V = f \times \lambda$

$$V = \frac{3000 \text{ m}}{\text{s}}$$

$$f = 150 \text{ kHz} = 150000 \text{ Hz}$$

$$\frac{3000 \text{ m}}{\text{s}} = 150000 \frac{\text{cycles}}{\text{s}} \times \lambda$$

$$\frac{3000 \text{ m}}{\cancel{8}} = \frac{150000 \text{ cycles} \times \lambda}{\cancel{8}} \\ \frac{150000 \text{ cycles}}{\cancel{8}} \quad \frac{750000 \text{ cycles}}{\cancel{8}}$$

$$\frac{.02 \text{ m}}{\text{cycle}} = \lambda$$

$$.02 \text{ m} \times \frac{1000 \text{ mm}}{1 \text{ m}} = \boxed{20 \text{ mm}}$$

6/10
Inverse Square Law (Safety)

$$Q45 \quad 35 \frac{\text{mR}}{\text{hr}} \times (35 \text{ ft})^2 = I_2 \times (180 \text{ ft})^2$$

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$$35 \frac{\text{mR}}{\text{hr}} \times 1225 \text{ ft}^2 = I_2 \times 32400 \text{ ft}^2$$

$$35 \frac{\text{mR}}{\text{hr}} \times 1225 \text{ ft}^2 = I_2 \times 32400 \text{ ft}^2$$

$$\frac{35 \times 1225}{32400} = I_2$$

$$1.32 \frac{\text{mR}}{\text{hr}} = I_2$$

Q47 For 1 inch thick, use

100 to 125 $\frac{\text{Amps}}{\text{inch}}$ spacing.

$$I_{\text{min}} = 5 \text{ inch} \times 100 \frac{\text{Amps}}{\text{inch}} = 500 \text{ Amps}$$

$$I_{\text{max}} = 5 \text{ inch} \times 125 \frac{\text{Amps}}{\text{inch}} = 625 \text{ Amps}$$

500 to 625 Amps is acceptable

7/10

Q 54

KRAFT • $\frac{50 \text{ mR}}{\text{hr}} \times 2 \text{ hr} = 100 \text{ mR}$

• Inverse square law.

$$\frac{50 \text{ mR}}{\text{hr}} \times D_1^2 = I_2 \times (2D)^2$$

$$\frac{50 \text{ mR}}{\text{hr}} \times D_1^2 = I_2 \times 4D^2$$

$$\frac{50 \text{ mR}}{\text{hr}} \times D_1^2 = I_2 \times 4D^2$$

$$\frac{50 \text{ mR}}{\text{hr}} = I_2$$

$$\frac{12.5 \text{ mR}}{\text{hr}} = I_2$$

• $\frac{12.5 \text{ mR}}{\text{hr}} \times 8 \text{ hrs} = 100 \text{ mR}$

• Total exposure = $100 \text{ mR} + 100 \text{ mR}$
= 200 mR

8/10

Q57 $1 \text{ ft} \cdot \text{c} = 10.761 \times$

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$$2 \text{ ft} \cdot \text{c} \times \frac{10.761 \times}{1 \text{ ft} \cdot \text{c}} = 21.521 \times$$

Q58 $\frac{500 \text{ mR}}{\text{hr} \cdot \text{Ci}} \times 10 \text{ Ci} = \frac{5000 \text{ mR}}{\text{hr}}$

$\nearrow$
14 ft

• Inverse square Law.

$$\frac{5000 \text{ mR}}{\text{hr}} \times (1 \text{ ft})^2 = I_2 \times (100 \text{ ft})^2$$

$$\frac{5000 \text{ mR}}{\text{hr}} \times 1 \text{ ft}^2 = I_2 \times 10000 \text{ ft}^2$$

$$\frac{5000 \text{ mR} \times 1 \text{ ft}^2}{10000 \text{ ft}^2} = \frac{I_2 \times 10000 \text{ ft}^2}{10000 \text{ ft}^2}$$

$$0.5 \text{ mR} = I_2$$

$$Q63 \quad \text{milli} = .001 \quad \frac{9}{10}$$

$$\text{KRAFT} \quad \text{Mega} = 1,000,000$$

001000000

$$-3 \text{ to } +6 = 9 \text{ orders of magnitude}$$

$$Q65 \quad \text{ft.c} = 10.76 \text{ ft}$$

Part D Miscellaneous

$$75. \quad \eta_{IO} = \frac{D_{oil}^2}{D_{IO}^2} = \frac{(10 \text{ mm})^2}{(15 \text{ mm})^2} = \frac{100 \text{ mm}^2}{225 \text{ mm}^2} = \underline{\underline{.444}}$$

or 44.4%

$$76. \quad \eta_{Enc} = \frac{D_{oil}^2}{D_{oil}^2} = \frac{(9 \text{ mm})^2}{(10 \text{ mm})^2} = \frac{81 \text{ mm}^2}{100 \text{ mm}^2} = \underline{\underline{.81}}$$

or 81%